Final Exam - EEE3094S 2020

Information

- The exam is open book.
- There are seven questions, totalling 100 marks. You must answer all of them.
- Submit your answers in pdf format. They may be typed or handwritten. Sketches may be hand-drawn and scanned, or constructed in a program such as PowerPoint. A .pptx file containing the plots has been provided.
- You must explain your reasoning and show your calculations for all questions.
- If you create or modify a plot to support your answer, ensure that you reference this graphic in your answer. You should also label the graphic with the question number. It is also helpful to annotate important features to draw attention to them.
- Closed-loop systems are in unity feedback unless otherwise stated.

Question 1

Figure 1.A shows the block diagram of a system that can be represented by the transfer function

$$T(s) = \frac{65(s+2)}{s^4 + 7s^3 + 32s^2 + 132s + 221}$$

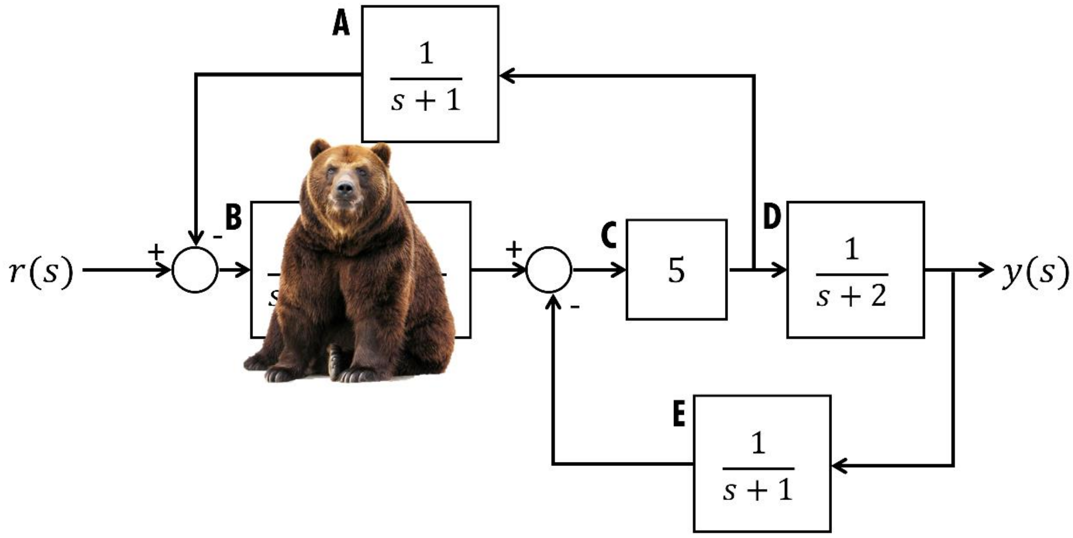


Figure 1.A – Block diagram.

A) [9 marks] Unfortunately, a bear is sitting in front of one of the blocks and refuses to move. You would prefer to resolve the situation peacefully, and luckily, your knowledge of block diagram algebra should allow you to do so. Use it to show that the transfer function of the obscured block must be $B(s) = \frac{13}{s^2+4s+13}$

B) [6 marks] The unit step response in Figure 1.B (on the next page) was produced by the hidden subsystem. With reference to the final value, overshoot, and settling time, show that it supports the model of $B(s)$ you worked out in part A.

You may find the following formulas useful:

Percentage overshoot: $100e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$ Peak time: $\frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$

where ζ is the damping ratio and ω_n is the natural frequency.

Total: 15 marks

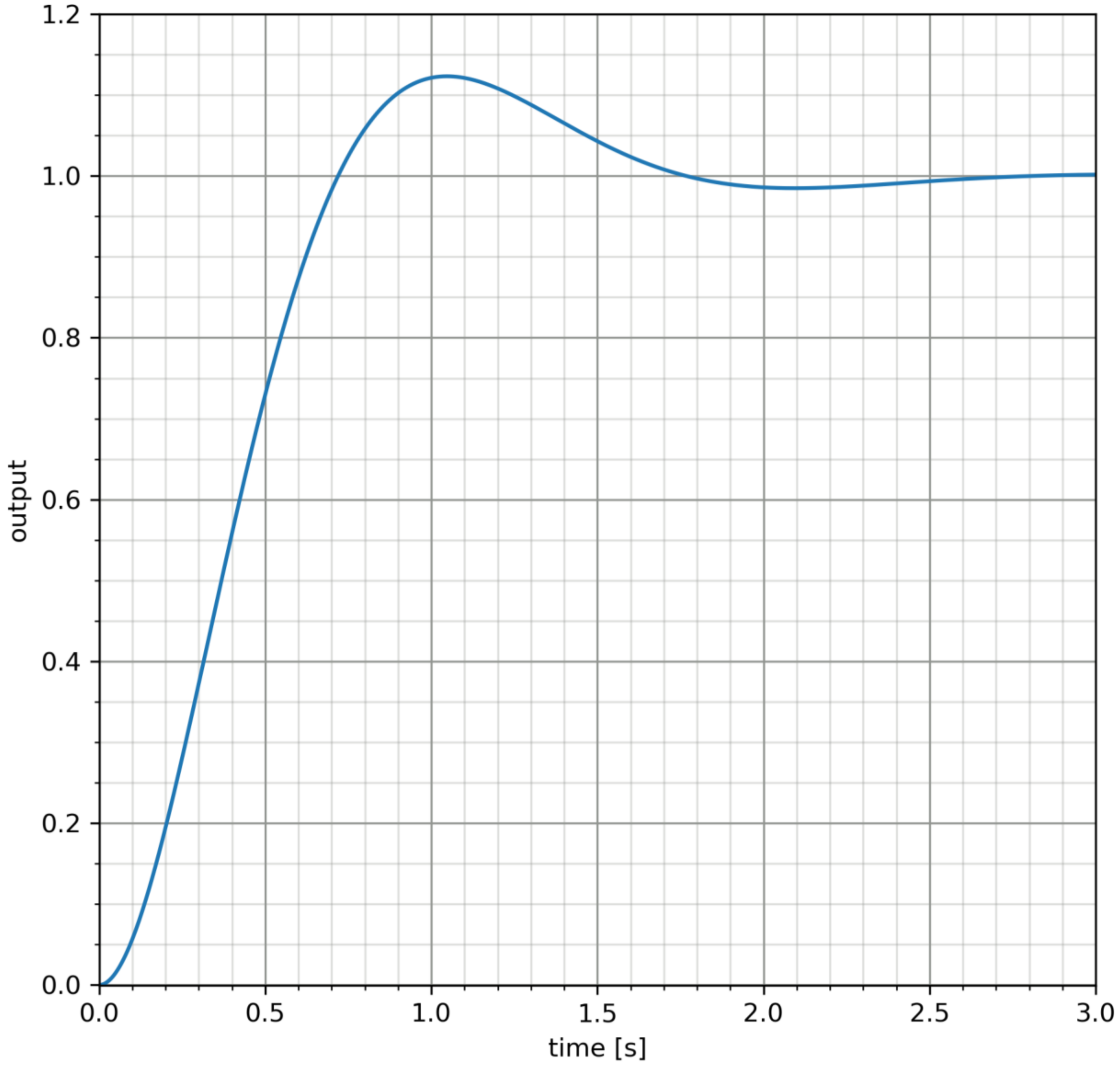


Figure 1.B – step response of $B(s)$

Question 2

Figure 2.A shows the block diagrams of three systems with the same type number.

A) [1 mark] What is the type number of these systems?

B) For each of the following design specifications, explain whether it can be met by all, none or some of the given systems.

i) [2 marks] Tracking a position setpoint without error
ii) [2 marks] Tracking a ramp setpoint without error
iii) [2 marks] Complete rejection of input disturbances in the form of a step
iv) [2 marks] Complete rejection of output disturbances in the form of a step

Total: 9 marks

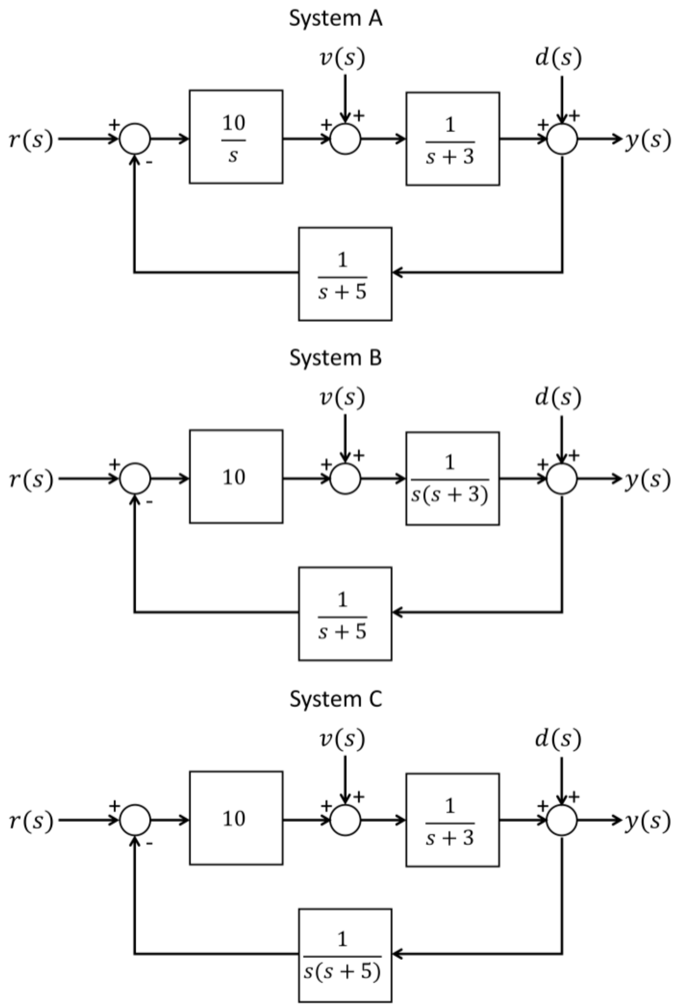


Figure 2.A – block diagrams of three systems. The setpoint, input disturbance and output disturbance are represented by the inputs $r(s), v(s)$ and $d(s)$, respectively.

Question 3

Decide whether each of the following statements are TRUE or FALSE. Explain your reasoning.

A) [3 marks] Consider the inverse Nichols chart shown in figure 3.A. If this system experiences an output disturbance, frequency components of the disturbance close to 0.75 rad/s will be magnified in the output of the closed-loop system.

B) [3 marks] The controller $C(s) = \frac{10(s-3)}{(s+5)}$ is a viable way to stabilize the unstable system $G(s) = \frac{1}{s-3}$ in closed loop.

C) [3 marks] The controller $C(s) = \frac{10(s+2)}{(s+10)}$ is a viable way to stabilize the unstable system $G(s) = \frac{1}{s-3}$ in closed loop.

D) [3 marks] The discrete-time system $G(z) = \frac{12z}{z-0.67}$ is stable.

E) [3 marks] The model $G_r(s) = \frac{1}{(s+2)(s+3)}$ provides a reasonably accurate reduced-order approximation of the third-order system $G(s) = \frac{1}{(s+2)(s+3)(s+15)}$.

Total: 15 marks

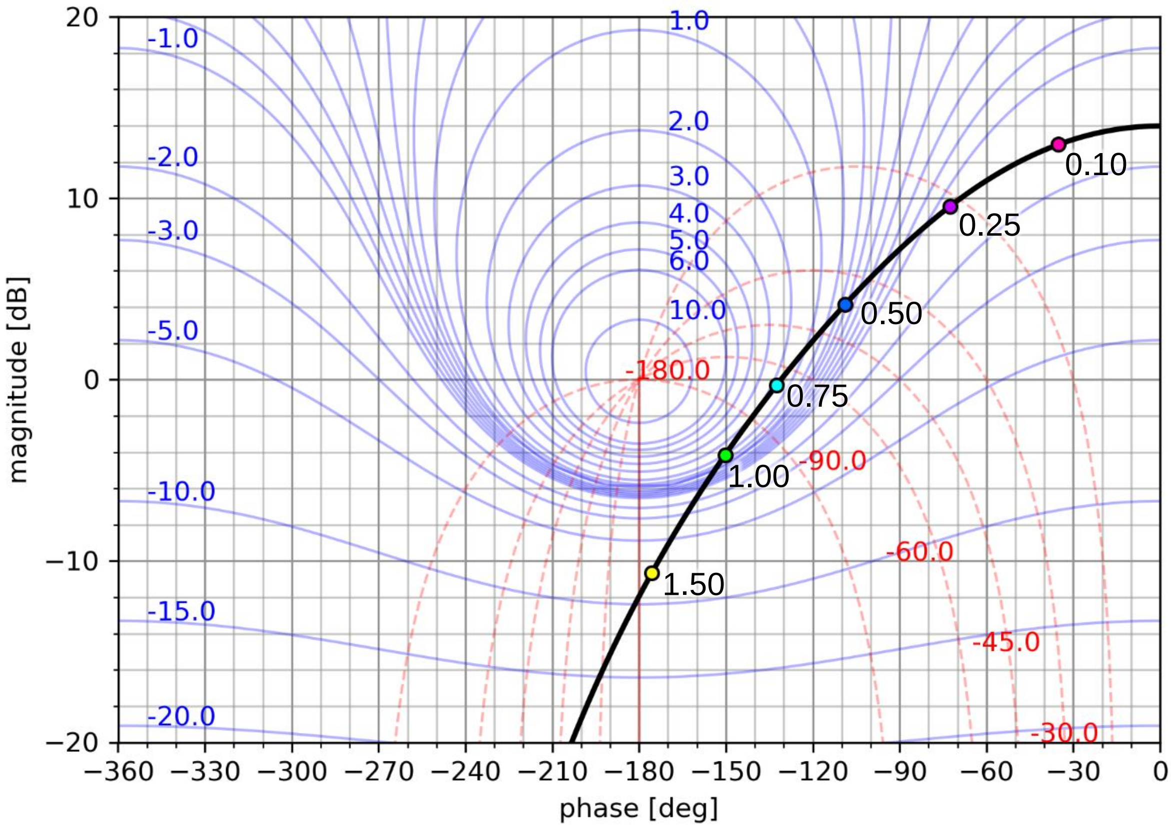
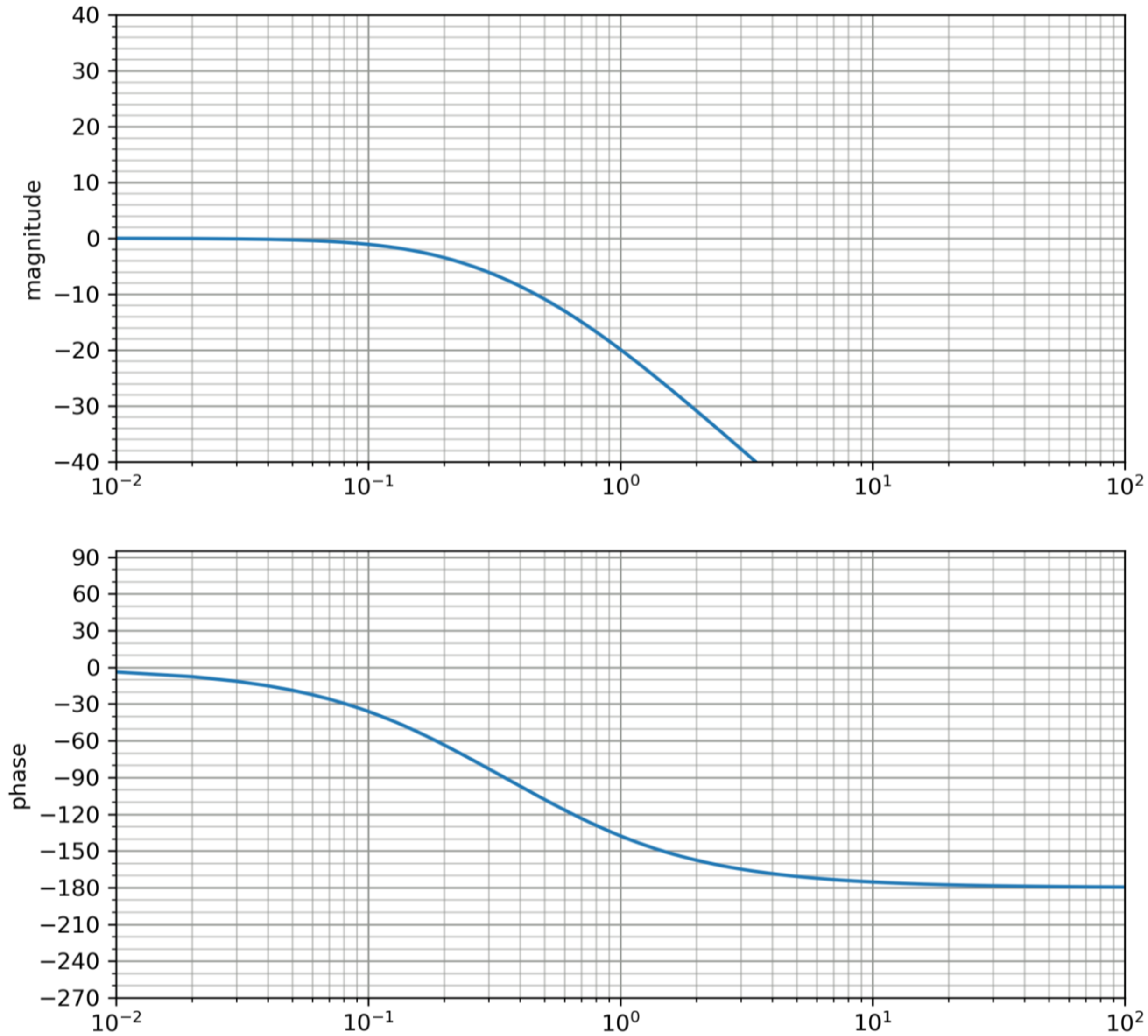


Figure 3.A – inverse Nichols chart. Annotated frequencies are in rad/s.

Question 4:

Figure 4.A shows the frequency response of a system G(s).



You are the head control engineer of a team that has been asked to design a lead compensator for this system. The requirements the design must satisfy include:

- Tracking a position input with >90% accuracy
- A phase margin of at least 60 degrees, because it is likely that the final controller will be implemented digitally.

Your colleagues present the following partially-completed designs for you to evaluate:

$$C_1(s) = K\frac{2.1\tau s + 1}{\tau s + 1}$$

Eevee

$$C_2(s) = K\frac{50\tau s + 1}{\tau s + 1}$$

Meowth

$$C_3(s) = K\frac{4\tau s + 1}{\tau s + 1}$$

Psyduck

A) [1 mark] Why is it necessary to increase the phase margin before implementing the controller digitally?

B)
   i) [2 marks] Show that a gain of $K = 10$ meets the accuracy specification and sketch the Bode plot of the system at this gain on the template in Figure 4.B.
   ii) [1 mark] What is the phase margin of the system at this gain?

C) [4 marks] Calculate the maximum phase that each design adds. Do you think that all the designs will be able to meet the phase margin specification?

D) [2 marks] Meowth's design might be difficult to implement using analogue components. Explain why, and suggest a modification to fix the problem that would still provide the same phase lead.

E) Psyduck is confused about how to proceed.
   i) [5 marks] Help him complete his design by calculating a suitable value of τ, and hence, the positions of the pole and zero of the compensator.
   ii) [5 marks] Sketch an approximate Bode plot for the compensator design, and combined system on the same plot. You may add your sketches to Figure 5.B, or use the empty template in 5.B. Confirm whether the design satisfies the phase margin specification.

Total: 20 marks

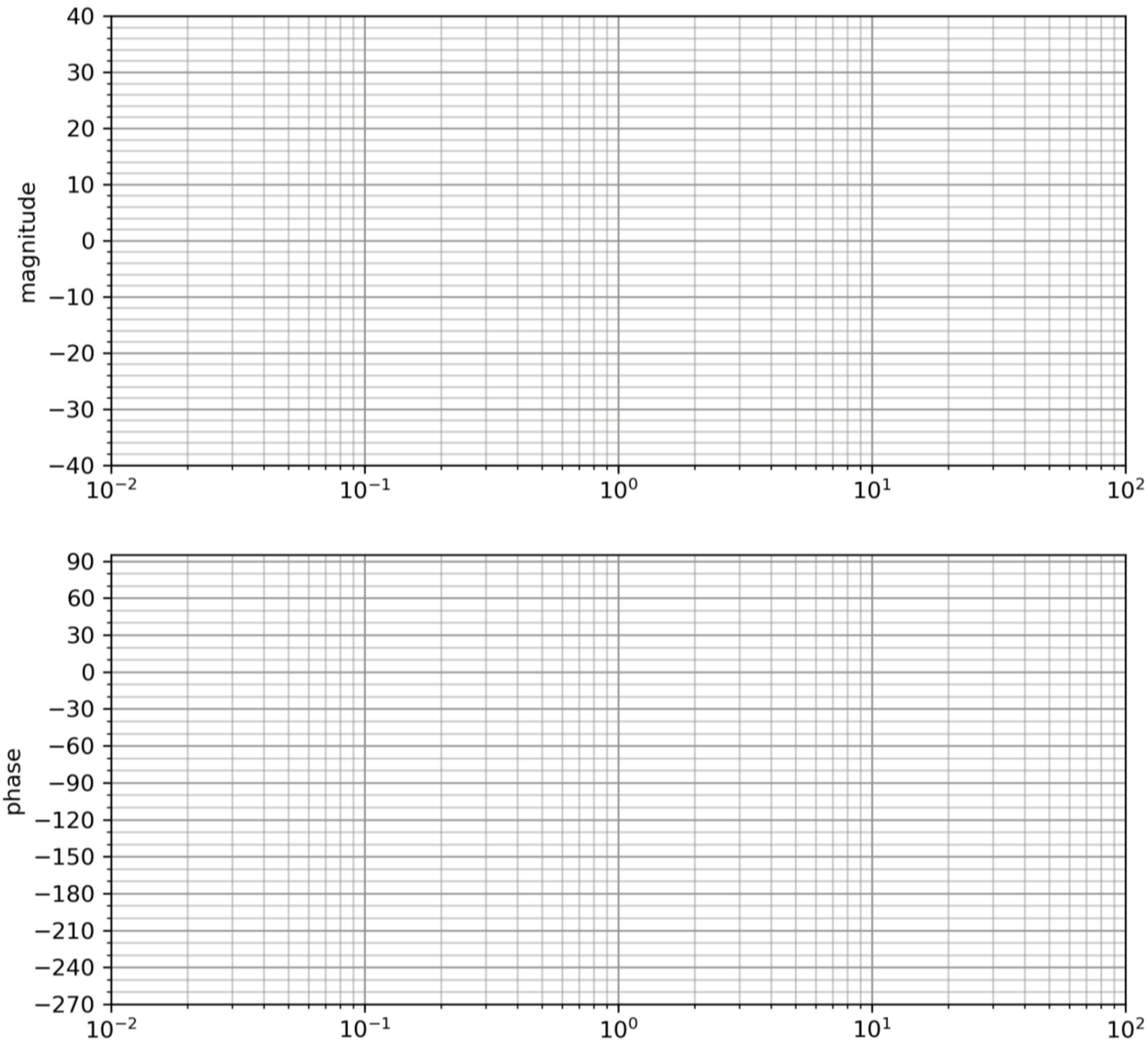


Figure 4.B – Bode plot template.

Question 5:

Consider the Nyquist plot in Figure 5.A (next page). This system is stable in open-loop.

A) Based on this plot, approximate the system's
   i) [2 marks] gain margin
   ii) [2 marks] phase margin
   iii) [2 marks] accuracy for tracking a step setpoint in closed loop (as a percentage)
   iv) [3 marks] damping ratio
   v) [6 marks] settling time to within 5% of its final value in closed loop.

B) This system is going to be used in an application where it will be subjected to a time delay of 0.2 seconds.
   i) [4 marks] Do you expect the closed-loop system to remain stable under these conditions?
   ii) [3 marks] What type of controller could you use to make the system more robust to delays? (You do not need to design the controller, just explain which of the controllers you have learned about would be suitable.)

Total: 20 marks

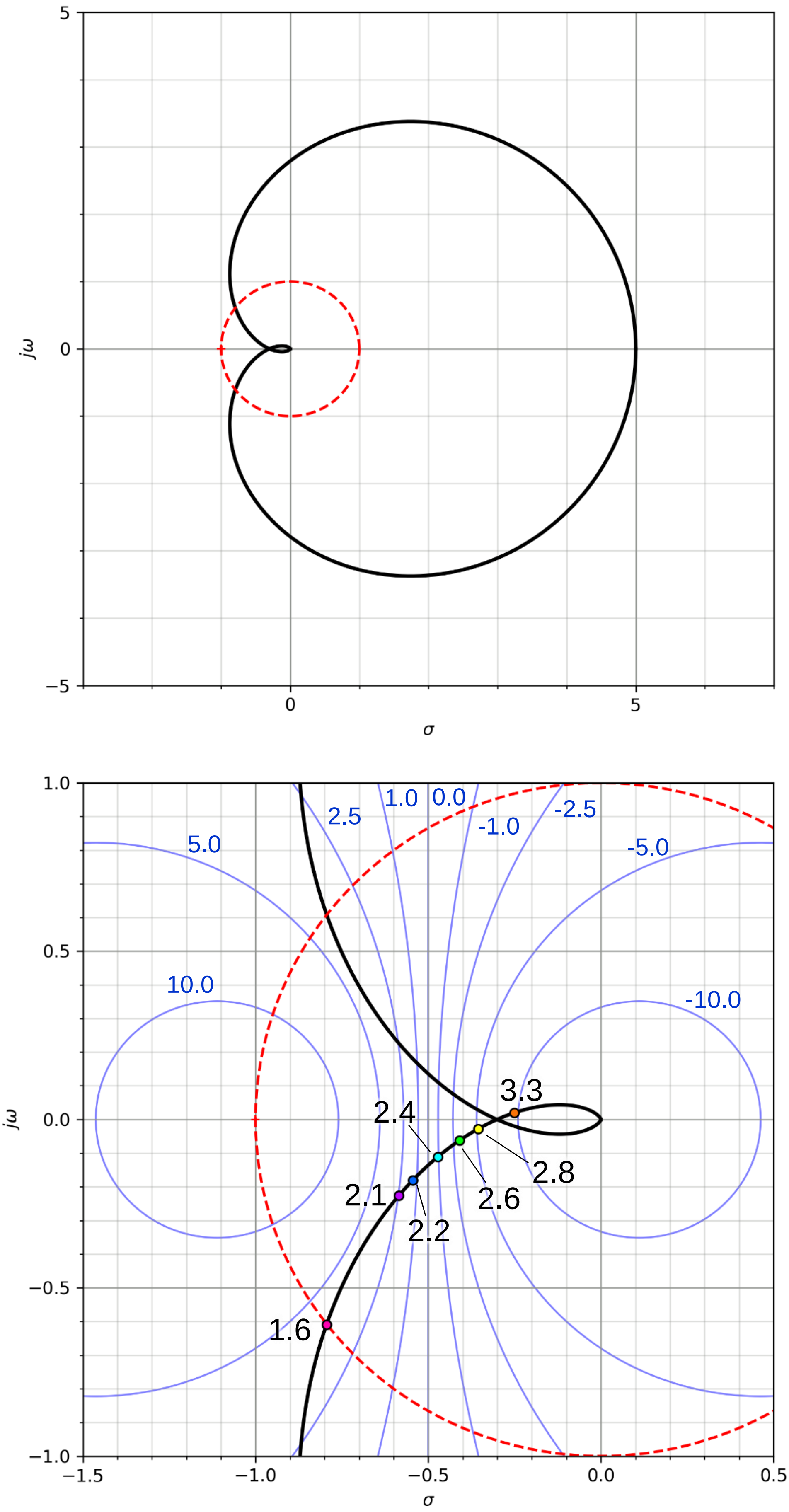


Figure 5.A Nyquist plot of system, and detailed view of Nyquist plot.

M circles are shown in blue. Their magnitudes are given in decibels.

The unit circle is shown in red.

Annotated frequencies are in rad/s

Question 6:

Figure 6.A shows the unit step responses of three systems, and Figure 6.B shows six pole-zero plots.

For each of the systems in Figure 6.A, suggest which of the pole-zero plots could have produced that response in open loop. (There might be more than one plausible option.)

Total: 6 marks

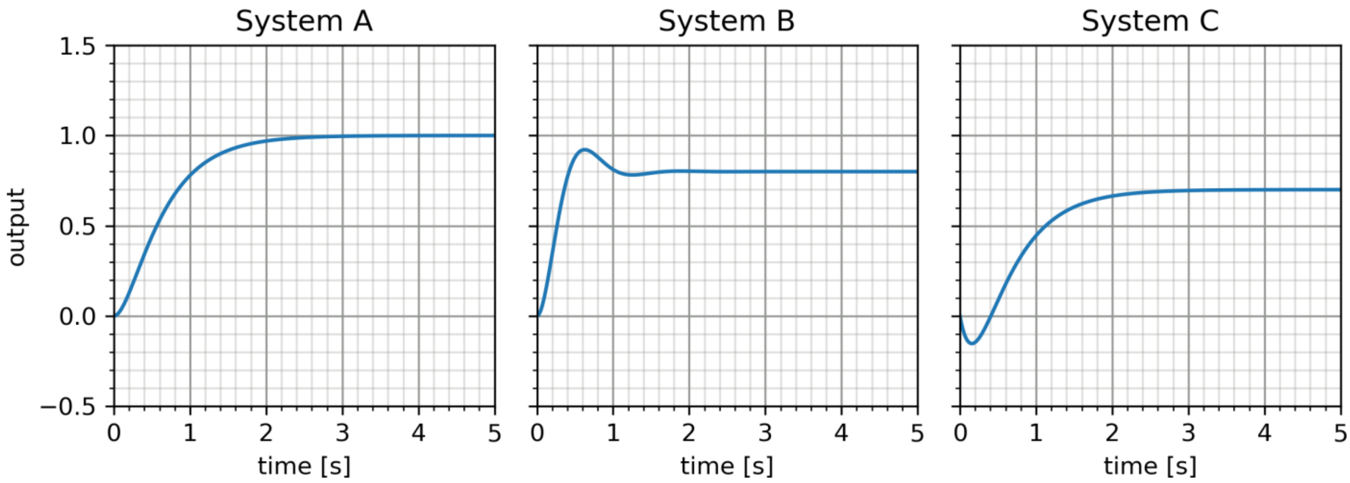


Figure 6.A – unit step responses

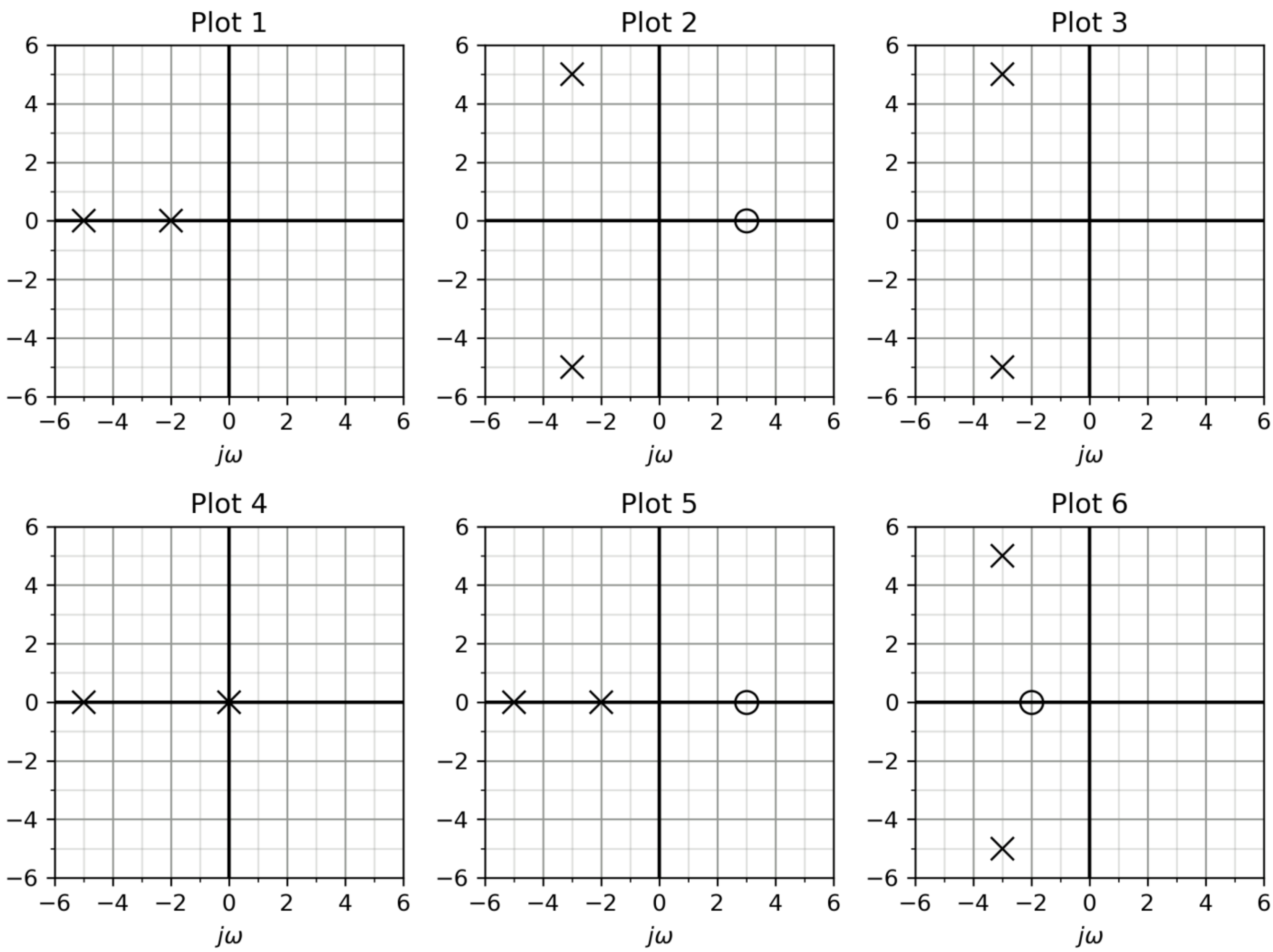


Figure 6.B – pole and zero plots

Question 7

Consider the system $G(s) = \frac{1}{(s+2)(s+4)}$

A) The system is connected in closed loop.
   i) [3 marks] What is the minimum possible closed-loop settling time (to within 2% of the final value) for this system? You may sketch a rough root locus plot to support your answer.
   ii) [3 marks] Calculate the gain required from a proportional controller to achieve this settling time and a damping ratio of 0.7.

B) The closed-loop system is required to track a position input with 95% accuracy.
   i) [2 marks] Calculate the error constant for a unit step input when the proportional controller you designed in part A.ii is included in the system.
   ii) [4 marks] Design a lag compensator to achieve the required accuracy. The combined system should have the same damping characteristics as in part A.ii.
   iii) [1 marks] Do you expect that the lag-compensated system will settle in more or less time than the value you worked out in part A.i?

C) [2 marks] A colleague suggests using a controller of the form $C(s) = \frac{K}{s}$ to satisfy the accuracy requirement instead. Briefly discuss the advantages and disadvantages of this idea.

Total: 15 marks